Learnbay

Masters In CS: Data Science & AI

Specialization in

Data Science & Analytics | Artificial Intelligence & ML

Detailed Syllabus

Foundation Track 10 ECTS Credits

Python Fundamentals

- Introduction To Python
- Datatypes
- Typecasting
- Operators
- Strings
- Conditional Statements
- Loops
- List
- Tuples
- Dictionary
- Functions
- Modules
- File Handling
- Iterators and Generators
- Exceptional Handling
- Regex
- Matplotlib
- Seaborn

Numerical Programming in python

- Numpy
- Pandas
- OOPS

Program Syllabus

Statistics & Machine Learning

20 ECTS Credits

Applied Statistics

- Introduction To Statistics
- Data Visualization Basics
- Descriptive Statistics
- Measure Of Symmetricity
- Z- Score Method
- Sampling
- Probability and Distribution
- Bayes Theorem
- Probability Mass Function
- Hypothesis Testing
- Hypothesis Test Method
- Error and ANOVA
- Linear Algebra

Advance machine learning

- Clustering
- PCA
- SVM
- Time Series

Problem-Solving Techniques

- Introduction to Problem Solving in Programming
- Understanding the Problem
- Pseudocode and Flowcharts
- Divide and Conquer
- Introduction to Data Science Problem Solving

Introduction to machine learning

- Introduction to Machine Learning
- Linear Regresssion
- Logistic Regression
- Decision Tree
- Cross validation
- KNN
- Naive Bayes

Problem-Solving Techniques(contd.)

- Introduction to Problem Solving in Programming
- Understanding the Problem
- Pseudocode and Flowcharts
- Divide and Conquer
- Introduction to Data Science Problem Solving
- Data Collection and Integration
- How to approach a Data Science Business Problem
- Different Types of Analysis
- Backtracking
- Debugging Techniques
- Machine Learning Pipelines

SPECIALZIATION 1

Data Science & Analtyics

Data Analytics & Visualization

20 ECTS Credits

Spreadsheets for Data Understanding - Advance Excel

- Basics of Excel
- Formatting a worksheet
- Adjusting row height & column width , changing cell alignment
- Managing your workbooks
- Advanced Printing options
- Editing a workbook
- Working with the forms menu
- Formulas
- Conditional Formatting
- Creating and working with charts
- Formatting a Data series and chart axis using fill effects
- Data analysis and pivot tables
- Automating tasks with macros
- Using excel events to triggers macros

Data visualization tools (Advance tableau, PowerBI, Looker)

Advance tableau

- Introduction to Tableau
- Adding Parameters
- Manipulating Graph size
- Funnel chart
- Actions
- Dashboard actions

Relational database - SQL fundamentals

- Introduction to SQL, RDBMS & DBMS
- Normalization & Constraints
- Working with joins & clause
- Views & Indexing
- Working with functions
- Subqueries & stored procedures
- Transaction, sequence & triggers

MongoDB

- Introduction to SQL, RDBMS & DBMS
- Normalization & Constraints
- Working with joins & clause
- Views & Indexing
- Working with functions
- Subqueries & stored procedures
- Transaction, sequence & triggers

Advance tableau

- Story point
- Connecting to sql server

Program Syllabus

Data Analytics & Visualization

20 ECTS Credits

Power BI

- Introduction & Installation
- Creating visuals
- Charts
- Advanced charts
- Power BI services
- Power query
- Power query continuation
- Data modelling

Advance machine learning

- Clustering
- PCA
- SVM
- Time Series

Google Looker

- Introduction to Looker
- Data modelling
- Creating queries & dashboards
- Data permission & Security in looker
- Data exploration with LookML
- Looker data blocks
- Advanced data modelling techniques

Business case studies

- Transportation safety domain
- HR domain
- Medical survey
- Sales
- Sports Analytics

Tools

tableau Power BI mongoDB. Looker

Program Syllabus

Deep Learning & Applied Analytics

20 ECTS Credits

Introduction to deep learning

- Introduction to Deep Learning
- MLP Architecture
- Weight Initialization
- Deep learning algorithms
- Exponentially Weighted Moving Average (EWMA)
- Padding
- Pooling
- Sequential API

Productionisation of ML System

- Introduction to ML Deployment
- ML Deployment Lifecycle
- Deployment Challenges
- Production Environment Overview
- Model Serialization
- Containerization Basics
- Containerizing ML Models
- Introduction to Cloud Platforms
- Cloud Deployment: AWS SageMaker
- Cloud Deployment: Azure ML Service
- Cloud Deployment: GCP AI Platform
- On-Premises Deployment
- Edge Deployment
- ML Model Monitoring
- A/B Testing in ML
- Continuous Integration & Continuous Deployment

Product Analytics

- Product Usage Onboarding
- A/B Testing Techniques
- Online Platform Metrics
- Customer Experience Mapping
- Marketing Performance Analysis
- Google Analytics for Beginners
- Google Analytics Reporting Techniques
- Data Pre-processing with Google Analytics

Productionisation of ML System

- Microservices for ML Deployment
- Kubernetes for ML Deployment
- Model Versioning
- Model Security & Privacy
- Optimizing Models for Deployment
- Batch vs. Real-time Inference
- Feedback Loops in ML Deployment
- Cost Management in ML Deployment
- Specialized Hardware for Deployment
- ML Ops & Best Practices
- Recap and Future Trends in ML Deployment
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Program Syllabus

Deep Learning & Applied Analytics

20 ECTS Credits

Generative AI

- Introduction to Generative AI
- Probabilistic Graphical Models
- Basic Generative Models
- Introduction to Generative Adversarial Networks (GANs)
- GAN Variants and Improvements - Part 1
- GAN Variants and Improvements - Part II
- Introduction to Variational Autoencoders (VAEs)
- Advanced VAE Variants
- Introduction to Transformer Models in Generative AI
- Overview of Autoregressive Models
- Neural Style Transfer
- Text-to-Image Synthesis
- Generative Models for Audio
- Ethical Implications and Challenges
- Advanced Topics and Recent Trends
- Project & Review

SPECIALZIATION 2

Artificial Intelligence & ML

Deep Learning & Advanced Data Analysis

20 ECTS Credits

Introduction to deep learning

- Introduction to Deep Learning
- MLP Architecture
- Weight Initialization
- Deep learning algorithms
- Exponentially Weighted Moving Average (EWMA)
- Padding
- Pooling
- Sequential API

Deep Learning for Natural Language Processing

- Introduction to NLP (Natural Language Processing)
- Natural Language Toolkit
- Stemming
- N-GRAMS
- Spacy
- Word2Vec
- NLP Spacy

Deep Learning For Computer Vision

- Introduction to Computer Vision
- Image Preprocessing
- Image Filtering and Enhancement
- Feature Detection and Extraction
- Image Segmentation
- Object Detection
- Object Tracking
- Image Classification:
- Image Recognition
- Semantic Segmentation
- Depth Estimation and 3D Vision
- Face Detection and Recognition:
- Pose Estimation
- Image Captioning
- Transfer Learning and Pretrained Models:
- Advanced Topics

High Dimensional Data Analysis

- Introduction to Big Data
- Data Locality
- SQOOP (SQL in HADOOP)
- Map Reduce
- Spark
- Persisting RDD

Program Syllabus

Gen-AI and Deployment

15 ECTS Credits

Generative AI

- Introduction to Generative AI
- Probabilistic Graphical Models
- Basic Generative Models
- Introduction to Generative Adversarial Networks (GANs)
- GAN Variants and Improvements - Part 1
- GAN Variants and Improvements - Part II
- Introduction to Variational Autoencoders (VAEs)
- Advanced VAE Variants
- Introduction to Transformer Models in Generative AI
- Overview of Autoregressive Models
- Neural Style Transfer
- Text-to-Image Synthesis
- Generative Models for Audio
- Ethical Implications and Challenges
- Advanced Topics and Recent Trends
- Project & Review

Productionisation of ML System

- Introduction to ML Deployment
- ML Deployment Lifecycle
- Deployment Challenges
- Production Environment Overview
- Model Serialization
- Containerization Basics
- Containerizing ML Models

Product Analytics

- Product Usage Onboarding
- A/B Testing Techniques
- Online Platform Metrics
- Customer Experience Mapping
- Marketing Performance Analysis
- Google Analytics for Beginners
- Google Analytics Reporting Techniques
- Data Pre-processing with Google Analytics

Productionisation of ML System (contd.)

- Introduction to Cloud Platforms
- Cloud Deployment: AWS SageMaker
- Cloud Deployment: Azure ML Service
- Cloud Deployment: GCP AI Platform
- On-Premises Deployment
- Edge Deployment
- ML Model Monitoring - Part 1
- ML Model Monitoring - Part II
- A/B Testing in ML
- Continuous Integration & Continuous Deployment
- CI/CD for ML - Part II
- Microservices for ML Deployment
- Kubernetes for ML Deployment
- Model Versioning

Program Syllabus

Gen-AI and Deployment

15 ECTS Credits

Productionisation of ML System (contd.)

- Real-world Deployment Case Studies
- Model Security & Privacy
- Optimizing Models for Deployment
- Batch vs. Real-time Inference
- Feedback Loops in ML Deployment
- Cost Management in ML Deployment
- Specialized Hardware for Deployment
- ML Ops & Best Practices
- Recap and Future Trends in ML Deployment

Program Syllabus

Comprehensive Data Structures Overview

10 ECTS Credits

Essential Data Structures: Lists, Queues, Stacks

- Introduction to Data Structures & Lists
- Array vs. Linked List
- Implementing a Singly Linked List - Part I
- Implementing a Singly Linked List - Part II
- Doubly Linked List - Basics
- Implementing a Doubly Linked List
- Circular Linked List
- Implementing a Circular Linked List
- Introduction to Stacks
- Stack Operations - Part I
- Stack Operations - Part II
- Applications of Stacks
- Evaluating Postfix Expressions Using Stacks
- Infix to Postfix Conversion Using Stacks
- Introduction to Queues
- Basic Queue Operations
- Circular Queues and their Implementation
- Queue Implementation Using Linked List
- Double-Ended Queue (Deque) Introduction
- Priority Queue Introduction
- Applications of Queues
- Comparison: Lists vs. Stacks vs. Queues

Essential Data Structures: Lists, Queues, Stacks (contd.)

- Common Problems and Solutions
- Recap and Advanced Resources

Advanced Data Structures: Trees, Graphs, Hash Tables

- Introduction to Trees
- Binary Trees
- Binary Tree Traversals
- Binary Search Trees (BST) - Part I
- Binary Search Trees (BST) - Part II
- Balanced Trees & AVL Trees - Part I
- Balanced Trees & AVL Trees - Part II
- Red-Black Trees - Part I
- Red-Black Trees - Part II
- B-Trees & B+ Trees
- Trie (Prefix Trees)
- Introduction to Graphs
- Representing Graphs in Memory
- Depth First Search (DFS)
- Breadth First Search (BFS)
- Shortest Path Algorithms - **Dijkstra's Algorithm**
- Shortest Path Algorithms - **Bellman-Ford Algorithm**
- Minimum Spanning Trees - **Prim's Algorithm**
- Minimum Spanning Trees - **Kruskal's Algorithm**

Program Syllabus

Comprehensive Data Structures Overview

10 ECTS Credits

Advanced Data Structures: Trees, Graphs, Hash Tables

- Graphs: Topological Sort
- Graphs: Strongly Connected Components
- Introduction to Hashing
- Collision Resolution - Part I
- Collision Resolution - Part II
- Dynamic Resizing in Hash Tables
- Advanced Hashing Techniques
- Applications of Hash Tables
- Trees: Range and Segment Trees
- Graphs: Network Flow and Maximum Flow Algorithm
- Advanced Hashing: Consistent Hashing
- Graphs: Traveling Salesman Problem
- Recap, Open Problems, and Future Study Directions

Need more questions answered? Reach out to us by clicking the button below

Enquire Now

Contact Us

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